

MAT 528 HW12

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Problem 1 (Munkres 58.2b). Determine whether the fundamental group of the torus T with a point removed is trivial, infinite cyclic, or isomorphic to the fundamental group of the figure eight.

Solution. Let T' be T with a point removed. Recall that T can be obtained by identifying opposite edges of a square, so T' can be obtained by identifying opposite edges of a square with a point x removed from its interior. This square-minus-a-point deformation retracts onto its boundary (specifically, for each p in the interior of the square, let q be the intersection of the ray from x through p with the boundary of the square, and move p linearly along the segment from p to q), so the image of the boundary under the quotient map identifying the edges of the square is a deformation retract of T' . On the other hand the boundary with its edges identified is simply the figure eight, so T' has the same fundamental group as the figure eight. $\square$

Problem 2 (Munkres 58.5). Recall that a space X is said to be **contractible** if the identity map from X to itself is nullhomotopic. Show that X is contractible if and only if X has the homotopy type of a one-point space.

Solution. Suppose first that X is contractible. Then there is a homotopy $H: X \times I \rightarrow X$ such that $H(x, 0) = x$ and $H(x, 1) = x_0$ for some fixed $x_0 \in X$ and all $x \in X$. Let i be the inclusion map $i: \{x_0\} \rightarrow X$ and let $f: X \rightarrow \{x_0\}$ be the map $f(x) = x_0$. Then $f \circ i: \{x_0\} \rightarrow \{x_0\}$ is (homotopic to) the identity on $\{x_0\}$ and $i \circ f: X \rightarrow X$ sends every point in X to x_0 , which via the homotopy H is homotopic to the identity on X ; this proves that f and i are homotopy equivalences between X and a one-point space.

Conversely, suppose X has the homotopy type of a one-point space Y and let $f: X \rightarrow Y$, $g: Y \rightarrow X$ be homotopy equivalences. Then $g \circ f: X \rightarrow X$ is the constant map sending every point in X to the unique point in the image of g and is homotopic to the identity on X by the assumption that f and g are homotopy equivalences. Thus the identity on X is nullhomotopic, i.e. X is contractible. $\square$

Problem 3 (Munkres 58.9abcd). Let b_0 be the point $(1, 0) \in S^1$; choose a generator γ for the infinite cyclic group $\pi_1(S^1, b_0)$. If $x_0 \in S^1$, choose a path $\alpha: [0, 1] \rightarrow S^1$ from b_0 to x_0 and define $\gamma_{x_0} = \hat{\alpha}(\gamma)$. Then γ_{x_0} generates $\pi_1(S^1, x_0)$. The element $\hat{\alpha}(\gamma)$ is independent of the choice of the path α , since the fundamental group of S^1 is abelian.

Now given $h: S^1 \rightarrow S^1$, choose $x_0 \in S^1$ and let $h(x_0) = x_1$. Consider the homomorphism $h_*: \pi_1(S^1, x_0) \rightarrow \pi_1(S^1, x_1)$. Since both groups are infinite cyclic, $h_*(\gamma_{x_0}) = d \cdot \gamma_{x_1}$ for some integer d (writing the group operation additively); this d is called the **degree** of h , denoted $\deg h$.

- (a) Show that d is independent of the choice of x_0 .
- (b) Show that if $h, k: S^1 \rightarrow S^1$ are homotopic, they have the same degree.
- (c) Show that $\deg(h \circ k) = (\deg h)(\deg k)$.
- (d) Compute the degrees of the constant map, the identity map, the reflection map $\rho(x_1, x_2) = (x_1, -x_2)$, and the map $h(z) = z^n$, where z is a complex number.

Solution. (a) Let $y_0 \in S^1$ and $y_1 = h(y_0)$. Let α_j be a path from x_j to y_j for $j = 1, 2$. Let $h_{*x}: \pi_1(S^1, x_0) \rightarrow \pi_1(S^1, x_1)$, $h_{*y}: \pi_1(S^1, y_0) \rightarrow \pi_1(S^1, y_1)$ be the induced homomorphisms by h . I claim that $\widehat{\alpha_1} \circ h_{*x} = h_{*y} \circ \widehat{\alpha_0}$. To see this, let g_{x_0} be a path in the path homotopy class γ_{x_0} . We have

$$h_{*y}(\widehat{\alpha_0}([g_{x_0}])) = h_{*y}([\overline{\alpha_0} \cdot g_{x_0} \cdot \alpha_0]) = [(h \circ \overline{\alpha_0}) \cdot (h \circ g_{x_0}) \cdot (h \circ \alpha_0)] = \widehat{h \circ \alpha_0}([h \circ g_{x_0}]).$$

Here $h \circ \alpha_0$ is a path from $h(x_0) = x_1$ to $h(y_0) = y_1$, as is α_1 . By Munkres's Exercise 52.3 (from HW10), $\widehat{\alpha_1} = \widehat{h \circ \alpha_0}$, so

$$h_{*y}(\widehat{\alpha_0}([g_{x_0}])) = \widehat{h \circ \alpha_0}([h \circ g_{x_0}]) = \widehat{\alpha_1}([h \circ g_{x_0}]) = \widehat{\alpha_1}(h_{*x}(g_{x_0})),$$

proving the claim (since $[g_{x_0}]$ generates $\pi_1(S^1, x_0)$). If $h_{*x}(\gamma_{x_0}) = d\gamma_{x_1}$, then $\widehat{\alpha_1}(h_{*x}(\gamma_{x_0})) = d\gamma_{y_1}$ by definition of $\gamma_\bullet$, while if $h_{*y}(\gamma_{y_0}) = d'\gamma_{y_1}$, then

$$h_{*y}(\widehat{\alpha_0}(\gamma_{x_0})) = h_{*y}(\gamma_{y_0}) = d'\gamma_{y_1},$$

but we proved above that $d\gamma_{y_1} = d'\gamma_{y_1}$, so $d = d'$, hence d is independent of the choice of basepoint.

(b) Fix $x_0 \in S^1$ and let $A: S^1 \times I \rightarrow S^1$ be a homotopy such that $A(x, 0) = h(x)$ and $A(x, 1) = k(x)$. If $g_{x_0} \in \gamma_{x_0}$ is a path in the path homotopy class γ_{x_0} , then $B(s, t) := A(g_{x_0}(s), t)$ is a homotopy (not path homotopy) between the paths $h(g_{x_0})$ and $k(g_{x_0})$.

If we consider S^1 as a subset of $\mathbb{C}$, then multiplying h and k by some fixed $z \in S^1$ does not change their degree, since all it does is change the basepoint of the paths when considering the induced homomorphisms. Thus, we can assume h and k both send x_0 to $1 \in \mathbb{C}$. In this case we can modify B to be a *path* homotopy between $h(g_{x_0})$ and $k(g_{x_0})$, as

$$C(s, t) := \frac{B(s, t)}{B(0, t)}$$

is still continuous (because the denominator is continuous in t and is never zero) and is now always equal to 1 when $s = 0, 1$. In sum, $h(g_{x_0})$ and $k(g_{x_0})$ are path homotopic, so they lie in the same homotopy class $\deg(h) \cdot \gamma_1 = \deg(k) \cdot \gamma_1$ in $\pi_1(S^1, 1)$, proving $\deg h = \deg k$.

(c) The degree of $h \circ k$ is the coefficient of $\gamma_{(h \circ k)(x_0)} = \gamma_{h(k(x_0))}$ in

$$(h \circ k)_*(\gamma_{x_0}) = h_*(k_*(\gamma_{x_0})) = h_*(\deg k \cdot \gamma_{k(x_0)}) = \boxed{\deg h \cdot \deg k} \cdot \gamma_{h(k(x_0))}.$$

(d) The identity map induces the identity isomorphism $\pi_1(S^1, x_0) \rightarrow \pi_1(S^1, x_0)$, so its degree is $\boxed{1}$ (γ_{x_0} gets sent to itself, $1 \cdot \gamma_{x_0}$).

The constant map sending all points in S^1 to some c induces the trivial homomorphism $\pi_1(S^1, x_0) \rightarrow \pi_1(S^1, c)$ since the image of each path under the constant map is constant. Thus, the degree is $\boxed{0}$ (γ_{x_0} gets sent to $0 \cdot \gamma_c$).

The reflection map sends a path α based at b_0 to $\bar{\alpha}$, so the induced homomorphism maps γ_{b_0} to $-\gamma_{b_0}$, hence its degree is $\boxed{-1}$.

The map $z \mapsto z^n$ sends loop α based at $b_0 = 1 \in \mathbb{C}$ to (a loop homotopic to) α concatenated with itself n times, namely

$$\beta(t) = \alpha(nt - \lfloor nt \rfloor)$$

where $\lfloor x \rfloor$ is the greatest integer that is at most x . Thus $h_*([\alpha]) = n[\alpha]$ implies that $h_*([\gamma_{b_0}]) = n\gamma_{b_0}$, so the degree of h is $\boxed{n}$. $\square$

Problem 4 (Munkres 59.3b). Show that $\mathbb{R}^2$ and $\mathbb{R}^n$ are not homeomorphic if $n > 2$.

Solution. If $\mathbb{R}^2$ and $\mathbb{R}^n$ are homeomorphic then the spaces obtained by removing one point from each space ought to be homeomorphic as well. However, by Munkres's Theorem 58.2, $\mathbb{R}^2$ minus a point has fundamental group isomorphic to that of S^1 , namely $\mathbb{Z}$, while for $n > 2$, $\mathbb{R}^n$ minus a point has fundamental group isomorphic to that of S^{n-1} , which is trivial by Munkres's Theorem 59.3. Homeomorphic spaces have the same fundamental group, so the inequality of the fundamental groups of S^1 and S^{n-1} for $n > 2$ implies that $\mathbb{R}^2$ and $\mathbb{R}^n$ cannot be homeomorphic. $\square$

Problem 5 (Munkres 60.1). Compute the fundamental groups of the “solid torus” $S^1 \times B^2$ and the product space $S^1 \times S^2$.

Solution. By Munkres's Theorem 60.1,

$$\begin{aligned}\pi_1(S^1 \times B^2, (x_0, y_0)) &\cong \pi_1(S^1, x_0) \times \pi_1(B^2, y_0) \cong \mathbb{Z} \\ \pi_1(S^1 \times S^2, (x_0, y_0)) &\cong \pi_1(S^1, x_0) \times \pi_1(S^2, y_0) \cong \mathbb{Z}\end{aligned}$$

since B^2 and S^2 both have trivial fundamental groups. $\square$

Problem 6 (Munkres 70.1). Suppose that the homomorphism i_* induced by the inclusion $i: U \cap V \rightarrow X$ is trivial.

(a) Show that j_1 and j_2 induce an epimorphism

$$h: (\pi_1(U, x_0)/N_1) * (\pi_2(V, x_0)/N_2) \rightarrow \pi_1(X, x_0),$$

where N_1 is the least normal subgroup of $\pi_1(U, x_0)$ containing the image of i_1 and N_2 is the least normal subgroup of $\pi_1(V, x_0)$ containing the image of i_2 .

(b) Show that h is an isomorphism.

Solution. Write U_1 and U_2 instead of U and V .

(a) Let $k \in \{1, 2\}$. Since $i_* = j_k \circ i_k$ is trivial, $\text{im } i_k \subseteq \ker j_k$, so $N_k \subseteq \ker j_k$. Letting $p_k: \pi_1(U_k, x_0) \rightarrow \pi_1(U_k, x_0)/N_k$ be the projection homomorphism, the fact that $N_k \subseteq \ker j_k$ means there exists a unique homomorphism $\bar{j}_k: \pi_1(U_k, x_0)/N_k \rightarrow \pi_1(X, x_0)$ such that $\bar{j}_k \circ p_k = j_k$. The map

$$h: (\pi_1(U_1, x_0)/N_1) * (\pi_2(U_2, x_0)/N_2) \rightarrow \pi_1(X, x_0)$$

induced by $\bar{j}_1$ and $\bar{j}_2$ has the same image as the map j of Munkres's Theorem 70.2 because $\bar{j}_k$ and j_k have the same image, and the aforementioned theorem tells us that j is an epimorphism, proving the desired result.

(b) Since h is an epimorphism it suffices to show that it has a left inverse. Let

$$\Phi: \pi_1(X, x_0) \rightarrow (\pi_1(U_1, x_0)/N_1) * (\pi_2(U_2, x_0)/N_2)$$

be the homomorphism given by taking $H = (\pi_1(U_1, x_0)/N_1) * (\pi_2(U_2, x_0)/N_2)$ in van Kampen's theorem (the maps $\phi_k: \pi_1(U_k, x_0) \rightarrow H$ are given by projecting $\pi_1(U_k, x_0)$ via p_k and then embedding the image $\pi_1(U_k, x_0)/N_k$ into H in the natural way).

It suffices to check that $\Phi(h(\alpha)) = \alpha$ for all αN_k in $\pi_1(U_k, x_0)/N_k$ (which is the coset of N_k by $\alpha \in \pi_1(U_k, x_0)$), since the free product is generated by the elements in the factors. For such an $\alpha N_k \in \pi_1(U_k, x_0)/N_k$, $h(\alpha) = \bar{j}_k(\alpha N_k) = j_k(\alpha)$ by construction. The commutative diagram shown in the theorem statement says that $\Phi \circ j_k = \phi_k$, so

$$\Phi(h(\alpha N_k)) = \Phi(j_k(\alpha)) = \phi_k(\alpha) = \alpha N_k$$

since, as described above, ϕ_k simply embeds the coset $p_k(\alpha) = \alpha N_k$ into H in the natural way. This shows that Φ is a left inverse of h as desired. $\square$